II. Cycles beyond 1 for the $qn + 1$ map: finiteness for each length, and an explicit bound on q for a fixed odd element

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Abstract

For the $qn + 1$ map, for each length at least 2, only finitely many odd integers q have a cycle of that length. The argument uses Ridout's theorem in the form that allows prime factors of the denominator outside a fixed finite set, and it gives no bound on q.

If a fixed odd positive integer lies on a cycle with two or three distinct odd numbers, then q lies below an explicit bound depending only on that integer. For three odd numbers, once q meets that bound, a lower bound and an upper bound cannot hold at the same time. Bauer and Bennett (2018) give an irrationality measure whose exponent, about 1.6149, lies just below the golden ratio, the threshold at which these bounds conflict. Every smaller exponent is a finite check. For two odd numbers the same conflict uses their exponent 1.465 for the square root of 2 (2002). When the fixed integer is 1 and the cycle has three odd numbers, the bound is $q < 2^{17990}$, and the search in the companion paper already rules out every such cycle.

1 Introduction

This paper studies cycles of the $qn + 1$ map for any starting value. It is preceded by the narrower paper [5], which studies cycles that pass through 1.

Let $q \geq 3$ be an odd integer.

For an odd positive integer n, let v be the largest integer such that 2^v divides $qn + 1$. The map C sends n to $(qn + 1)/2^v$. A cycle of length ℓ is ℓ distinct odd positive integers, each sent to the next by C, and the last sent back to the first. The integer h is the total number of factors of 2 removed along the cycle.

The complete list of known cycles is short.

A length-1 cycle is a fixed point: C sends some odd positive integer x_0 to itself, hence $x_0(2^h - q) = 1$. The only positive odd solution is $x_0 = 1$ and $q = 2^h - 1$, so $1 \to 1$ when $q = 2^e - 1$ with $e \geq 2$, and there is no length-1 cycle with $x_0 > 1$. For $q = 5$ and $h = 5$ there is $1 \to 3 \to 1$. For $q = 5$ and $h = 7$ there are $13 \to 33 \to 83 \to 13$ and $17 \to 43 \to 27 \to 17$. For $q = 181$ and $h = 15$ there are $27 \to 611 \to 27$ and $35 \to 99 \to 35$. Crandall recorded the cycle $27 \to 611$ from the identity $2^{15} - 181^2 = 7$.

Conjecture 1. These are all the cycles of C. For every odd $q \geq 3$, every cycle is either $1 \to 1$ with $q = 2^e - 1$, or one of the five cycles at $q = 5$ and $q = 181$ listed above. In particular, if $q \geq 7$ is neither 181 nor of the form $2^e - 1$, then C has no cycle, so every orbit diverges.

There is then no cycle through 1 other than $1 \to 1$ for Mersenne q and $1 \to 3 \to 1$ for $q = 5$, which is the conjecture of Santos (2020), and for $q = 3$ there is no cycle other than $1 \to 1$. For those $q \geq 7$ that are neither 181 nor Mersenne, the claim is stronger than Crandall's conjecture that some orbit avoids 1: there is no cycle at all, and an infinite orbit of odd positive integers is unbounded. The searches that found no others, in the ranges they cover, are in the companion paper [5]. Section 8 gives a heuristic for the expected number of cycles.

Theorem 2. For each $\ell \geq 2$, only finitely many odd q have a cycle of C of length ℓ.

Theorem 2 is ineffective: it rests on the general form of Ridout's theorem [6] in Section 3, which allows prime factors of the denominator outside a fixed finite set, and gives no bound on q, and no bound depending on a fixed starting value. (A finiteness result is called effective if its proof gives an explicit bound on any exception, so that a finite computation can find them all, and ineffective if it gives no such bound.) The case in which the cycle visits 1 is Theorem 3 of the companion paper, and that proof uses only the S-unit form of Ridout's theorem. The proof of Theorem 2 is in Section 3.

The argument of Theorem 3 is the comparison already written in the proof of Theorem 2 of [5]. That theorem assumes the cycle visits 1, and that the odd part of each denominator is 1 or ℓ. Here the cycle need not visit 1. The odd part is x, or $3x$ in the length-3 product. The exponent is still below the golden ratio $\varphi = (1 + \sqrt{5})/2$, so the same comparison forces q below a bound read off from x.

Theorem 3. Let x be an odd positive integer, and let $Q(x) = \max(Q_2(x), Q_3(x))$ be the integer defined in Section 5. If x lies on a cycle of C of length 2 or of length 3, then $q < Q(x)$. A length-2 cycle satisfies the stronger $q < Q_2(x)$. A length-3 cycle satisfies the stronger $q < Q_3(x)$.

For $x = 1$ and length 3, Section 5 gives $Q_3(1) < 2^{17990}$. Any such cycle has one candidate $q = \lfloor 2^{h/3} \rfloor$ for each h, so it is enough to check $h < 53970$. The companion paper checks every $h \leq 200{,}000$ and finds no cycle. That search includes $h < 53970$, and Theorem 1 of [5] is the resulting emptiness. Length 2 through 1 is Proposition A.1 of [5]: the only such q is 5.

The bound is not an emptiness result for a general x. At $q = 181$ the cycles $27 \to 611 \to 27$ and $35 \to 99 \to 35$ both occur. Section 5 records $181 < Q_2(27)$ and $181 < Q_2(35)$.

Section 6 takes up cycles of length 2 with no element fixed in advance. The three known cycles are the only ones with $D = 7$. Every other cycle of length 2 has $q \geq 2^{4500}$ and $t \geq 2409$.

Length 4 with h odd remains open, as in the companion paper: the roots are $2^{1/4}$ and $2^{3/4}$, and a fixed odd part does not supply an exponent below φ for them. Corollary 4 covers the longer classes whose roots are only $\sqrt{2}$, or only the cube roots of 2.

2 Definitions and notation

Throughout, $q \geq 3$ is an odd integer and C is the map defined in Section 1.

The 2-adic valuation. For a nonzero integer n, $v_2(n)$ is the largest integer v such that 2^v divides n.

Cycles. A cycle of C of length $\ell \geq 2$ consists of ℓ distinct odd positive integers. If those integers are $x_0, x_1, \ldots, x_{\ell-1}$, then $C(x_{i-1}) = x_i$ and $C(x_{\ell-1}) = x_0$, where the indices are read modulo ℓ.

Halvings. The halving counts $k_1, \ldots, k_\ell$ of the cycle are defined by

$$qx_{i-1} + 1 = 2^{k_i} x_i \qquad (1 \leq i \leq \ell,\ x_\ell = x_0),$$

so $k_i = v_2(q \cdot x_{i-1} + 1)$. The total number of halvings is $h = k_1 + \cdots + k_\ell$, and $S_i = k_1 + \cdots + k_i$.

The golden ratio. The golden ratio is $\varphi = (1 + \sqrt{5})/2$.

The length-3 constants. Let $N = 16166467234$. The exponent μ_3 is $3 \log_2 N/63$. The positive number γ_3 is $1 - \mu_3 + 1/\mu_3$. For a positive integer s', the constant $c_3(s')$ is $2^{-5}\, N^{-79/63}/s'$.

Odd parts at length 3. Let x be an odd positive integer. The positive integer s'_{near} is $x/\gcd(x, 3)$. The positive integer s'_{win} is x.

3 Ineffective finiteness

Ridout's theorem (1958). Let α be an algebraic irrational, let S be a finite set of primes, and let $\varepsilon > 0$. Only finitely many reduced fractions p/Q with $Q > 0$ have every prime factor of Q in S and satisfy

$$\left|\alpha - \frac{p}{Q}\right| < Q^{-1-\varepsilon}.$$

The same theorem has the general form: writing $|n|_p = p^{-v_p(n)}$ for a nonzero integer n, only finitely many rationals P/Q in lowest terms satisfy

$$Q|Q\alpha - P| \prod_{p \in S} |Q|_p < \max(|P|, Q)^{-\varepsilon}.$$

The first statement is the S-unit form. If every prime factor of Q lies in S, then $\prod_{p \in S} |Q|_p = 1/Q$ and the left side of the general form equals $|Q\alpha - P|$. Whenever $|P| \leq Q$, this is the first form. The general form allows prime factors of Q outside S. Such a prime does not appear in the product, and it contributes its full size to the height $\max(|P|, Q)$. Neither form gives a bound on the exceptional fractions. Since α is irrational, the finitely many fractions that meet the general form for a fixed α lie at a positive distance from α, and these distances have a positive minimum. This is why Theorem 2, which uses the general form in place of an effective measure, gives no bound on q.

Lemma 1 (the window for any cycle). Let $a = x_0 \to x_1 \to \cdots \to x_{\ell-1} \to a$ be a cycle of length $\ell \geq 2$, with a its smallest element, and let $\sigma = \sum_{i=0}^{\ell-1} 1/x_i$. Then

$$2^h = \prod_{i=0}^{\ell-1} \left(q + \frac{1}{x_i} \right),$$

so $q^\ell < 2^h < (q+1)^\ell$, and in particular $\ell \nmid h$ and $q = \lfloor 2^{h/\ell} \rfloor$. Writing $2^{h/\ell} = q + \delta$,

$$e^{-\ell/q} \leq \frac{\ell\,\delta}{\sigma} \leq e^{\ell/q}.$$

Proof. Multiplying $2^{k_i} x_i = q\,x_{i-1} + 1$ around the cycle gives the product. Each factor lies in $(q,\, q+1]$, with equality to $q+1$ only for an element equal to 1. At most one element is 1, and $\ell \geq 2$, so the product is strictly larger than q^ℓ and strictly smaller than $(q+1)^\ell$. A power of 2 strictly between consecutive ℓ-th powers is not an ℓ-th power, so $\ell \nmid h$, and $q = \lfloor 2^{h/\ell} \rfloor$. In particular $0 < \delta < 1$.

The binomial theorem gives

$$\ell q^{\ell-1}\delta \leq (q+\delta)^\ell - q^\ell \leq \ell q^{\ell-1}\delta \Big(1 + \frac{\delta}{q}\Big)^{\ell-1}.$$

Dividing the binomial expansion by $\ell q^{\ell-1}\delta$, the coefficient of $(\delta/q)^k$ is $\binom{\ell}{k+1}/\ell = \binom{\ell-1}{k}/(k+1) \leq \binom{\ell-1}{k}$, and $\sum_{k\geq 0} \binom{\ell-1}{k}(\delta/q)^k = (1+\delta/q)^{\ell-1}$. Since $\delta < 1$, this factor is at most $e^{(\ell-1)/q} \leq e^{\ell/q}$.

The same difference equals $\sum_{j\geq 1} e_j q^{\ell-j}$, where e_j is the j-th elementary symmetric function of the reciprocals $1/x_i$. Thus $e_1 = \sigma$, and the right side is at least $q^{\ell-1}\sigma$. In the expansion of σ^j, each product of j distinct reciprocals occurs $j!$ times, so $e_j \leq \sigma^j/j!$. Hence the right side is at most

$$\sum_{j\geq 1} \frac{\sigma^j}{j!}\, q^{\ell-j} = q^{\ell-1} \sum_{m\geq 0} \frac{\sigma^{m+1}}{(m+1)!\, q^m} \leq q^{\ell-1}\sigma\, e^{\sigma/q},$$

where the last step uses $(m+1)! \geq m!$. Every element is at least 1, so $\sigma \leq \ell$ and $e^{\sigma/q} \leq e^{\ell/q}$. Comparing the two sides gives $\ell\delta \leq \sigma\, e^{\ell/q}$ and $\ell\delta\, e^{\ell/q} \geq \sigma$. □

Lemma 2 (three approximations). In the setting of Lemma 1, write $h = \ell m + r$ with $0 < r < \ell$. Let $\theta_i = 2^{ir/\ell}$ for $1 \leq i \leq \ell - 1$, let $\theta = \theta_1$, and let $y = \min(x_1, \ldots, x_{\ell-1})$. Suppose $q \geq 4\ell$. Then $2^m > q/2$, and:

1. $|\theta - q/2^m| = \delta/2^m$.
2. For $1 \leq i \leq \ell - 1$, the rational number $F_i = x_i/(a \cdot 2^{im - S_i})$ satisfies $|\theta_i - F_i| \leq 2i\theta_i/(aq)$.
3. The rational number $W = (\ell aq + 1)/(\ell a \cdot 2^m)$ satisfies $|\theta - W| \leq 2(1/y + 2\ell/q)/q$.

Proof. Since $2^m\theta = q + \delta$ and $\theta = 2^{r/\ell} < 2$, one has $2^m = (q+\delta)/\theta > q/2$, and part 1 follows.

The relation $q\,x_{i-1}+1=2^{k_i}x_i$ gives $x_i\,2^{S_i}=a\prod_{u<i}(q+1/x_u)$ by induction on i, so

$$F_i=\frac{1}{2^{im}}\prod_{u<i}\Big(q+\frac{1}{x_u}\Big),\qquad \theta_i=\frac{(q+\delta)^i}{2^{im}}.$$

Lemma 1 gives $\delta\le(\sigma/\ell)\,e^{\ell/q}$. Every element is at least a, so $\sigma\le\ell/a$ and $\delta\le c/a$ with $c=e^{\ell/q}\le e^{1/4}$. Each factor $q+1/x_u$ and the number $q+\delta$ therefore lie in $[q,\,q+c/a]$, and both numerators lie in $[q^i,\,(q+c/a)^i]$. Their difference is at most

$$\Big(q+\frac{c}{a}\Big)^i-q^i=q^i\Big(\Big(1+\frac{c}{aq}\Big)^i-1\Big)\le\frac{ic}{a}\,q^{i-1}\,e^{ic/(aq)},$$

where $e^z-1\le z\,e^z$. Dividing by $2^{im}=(q+\delta)^i/\theta_i\ge q^i/\theta_i$ gives $|\theta_i-F_i|\le(ic\theta_i/(aq))\,e^{ic/(aq)}$. Here $i\le\ell-1$ and $aq\ge4\ell$, so $ic/(aq)<c/4$. For $0<z\le1/3$ and $n\ge2$, one has $n!\ge2^{n-1}$, so $z^n/n!\le z^n/2^{n-1}$ and $\sum_{n\ge2}z^n/2^{n-1}=z^2/(2-z)$. Thus $e^{1/4}<1+1/4+(1/16)/(7/4)=9/7<4/3$ and $e^{1/3}<1+1/3+1/15=7/5$. Since $c\le e^{1/4}<4/3$, one has $c/4<1/3$ and $c\,e^{c/4}<(4/3)\cdot(7/5)=28/15<2$. Thus $|\theta_i-F_i|\le2i\theta_i/(aq)$.

For part 3,

$$|\theta-W|=\frac{1}{2^m}\Big|\delta-\frac{1}{\ell a}\Big|.$$

Write $\delta-1/(\ell a)=(\sigma-1/a)/\ell+(\delta-\sigma/\ell)$. The other elements are at least y, so $0\le\sigma-1/a\le(\ell-1)/y$ and $(\sigma-1/a)/\ell\le1/y$. From the two bounds in Lemma 1, with $z=\ell/q$,

$$\frac{\sigma}{\ell}-\delta\le\frac{\sigma}{\ell}(1-e^{-z}),\qquad \delta-\frac{\sigma}{\ell}\le\frac{\sigma}{\ell}(e^z-1).$$

The inequality $1-e^{-z}\le e^z-1\le z\,e^z$ and $\sigma\le\ell$ give $|\delta-\sigma/\ell|\le(\ell/q)\,e^{\ell/q}$. Since $q\ge4\ell$, this is at most $(\ell/q)\,e^{1/4}<2\ell/q$. Therefore $|\delta-1/(\ell a)|\le1/y+2\ell/q$. Dividing by $2^m>q/2$ gives part 3. □

Theorem 2. For each $\ell\ge2$, only finitely many odd q have a cycle of C of length ℓ.

The argument does not produce a bound on q, and it does not produce a bound depending on a fixed odd start.

Proof of Theorem 2. Fix $\ell\ge2$, and set $\varepsilon=1/(10\ell)$. Let S consist of 2 and the prime factors of ℓ, and let Θ be the set of real numbers $2^{t/\ell}$ with $1\le t\le\ell(\ell-1)$ and $\ell\nmid t$. Each element of Θ is an algebraic irrational. Apply the general form of Ridout's theorem to each $\theta'\in\Theta$, with this S and ε. Only finitely many fractions meet the inequality, and none equals θ', so there is an $\eta>0$ such that every such fraction satisfies $|\theta'-P/d|\ge\eta$. The value of η is not provided by Ridout's theorem. The odd $q<4\ell$ are finite in number, so take an odd $q\ge4\ell$ with a cycle of length ℓ, in the notation of Lemmas 1 and 2. Then $\theta\in\Theta$. Each θ_i either lies in Θ or is a power of 2.

Consider the fraction $q/2^m$, which is in lowest terms. Its left side in the general form equals $2^m\cdot\delta\cdot2^{-m}=\delta$, since $|2^m|_p=1$ for every odd prime p. Also $\max(q,\,2^m)<q+1$, because $2^m=(q+\delta)/\theta<(q+1)/\theta<q+1$. If $\delta<(q+1)^{-\varepsilon}$, then $\delta<\max(q,\,2^m)^{-\varepsilon}$, so Ridout's inequality applies and $\delta/2^m\ge\eta$. Combined with $2^m>q/2$ this yields $q<2/\eta$.

Now assume $\delta \ge (q+1)^{-\varepsilon}$. Lemma 1 gives $\delta \le e^{\ell/q}/a \le e^{1/4}/a$, so $a < 2(q+1)^{\varepsilon}$.

Suppose first that $y \le q^{1-2\varepsilon}$, and choose i with $x_i = y$. If ℓ divides ir, then $\theta_i = 2^s$ for some integer $s \ge 1$. This power is not equal to F_i. Otherwise $x_i = a \cdot 2^{im - S_i + s}$; the right side is an odd positive integer only if the exponent of 2 vanishes, which forces $x_i = a$, contradicting distinctness. Let $e = \max(im - S_i,\, 0)$. In lowest terms the denominator of F_i divides $a \cdot 2^e$, so $|\theta_i - F_i| \ge 1/(a \cdot 2^e)$. Lemma 2 then gives $2^e \ge q/(2i\theta_i)$. If $e = 0$, then $q \le 2i\theta_i$. Here $i \le \ell - 1$ and $\theta_i < 2^{\ell-1}$, so $q < (\ell-1)\,2^{\ell}$. If $e \ge 1$, the same lemma and $q \ge 4\ell$ give $|\theta_i - F_i| \le 2(\ell-1)\theta_i/(4\ell) < \theta_i/2$, hence $F_i > \theta_i/2$. Therefore

$$y = F_i \cdot a \cdot 2^e > \frac{\theta_i}{2} \cdot a \cdot \frac{q}{2i\theta_i} = \frac{aq}{4i} \ge \frac{q}{4(\ell-1)}.$$

Once $q > (4(\ell-1))^{5\ell}$, the right side exceeds $q^{1-2\varepsilon}$, because $1/(2\varepsilon) = 5\ell$. This contradicts $y \le q^{1-2\varepsilon}$.

If instead ℓ does not divide ir, then $\theta_i \in \Theta$. Write $F_i = P/d$ in lowest terms. The odd part of d divides a, so $d \cdot |d|_2 \le a$. Every factor $|d|_p$ is at most 1 and $2 \in S$, so the left side of the general form is at most

$$d^2 \cdot |\theta_i - F_i| \cdot |d|_2 \le \frac{2i\theta_i\, d}{q}.$$

As above, $|\theta_i - F_i| < \theta_i/2$, so $F_i > \theta_i/2 > 1/2$ and $F_i < (3/2)\theta_i < 2^{\ell}$. If $im \ge S_i$, the fraction before reducing is $y/(a \cdot 2^{im - S_i})$, whose denominator equals $y/F_i < 2y$, and the numerator is y. If $im < S_i$, that denominator is a, so $d \le a \le y$, and the numerator is $F_i d < 2^{\ell} y$. In both cases $d < 2y$ and $\max(|P|,\, d) < 2^{\ell} y$. Also $2i\theta_i < (\ell-1)\,2^{\ell}$, so the left side is less than $(\ell-1)\,2^{\ell+1} y/q$. The right side is greater than $(2^{\ell} y)^{-\varepsilon}$. The comparison

$$(\ell-1)\,2^{\ell+1}\, y^{1+\varepsilon} < q \cdot 2^{-\ell\varepsilon}$$

therefore implies Ridout's inequality. It holds for every admissible y once

$$q^{\varepsilon + 2\varepsilon^2} > (\ell-1)\,2^{\ell+1+\ell\varepsilon},$$

because $y^{1+\varepsilon} \le q^{(1-2\varepsilon)(1+\varepsilon)} = q^{1-\varepsilon-2\varepsilon^2}$. The right side depends only on ℓ. For every larger q, Ridout's theorem gives $|\theta_i - F_i| \ge \eta$. Lemma 2 gives $2i\theta_i/(aq) \ge \eta$, and $2i\theta_i < (\ell-1)\,2^{\ell}$ with $a \ge 1$, so $q < (\ell-1)\,2^{\ell}/\eta$.

Suppose instead that $y > q^{1-2\varepsilon}$. The numerator $N = \ell a q + 1$ is congruent to 1 modulo ℓa, so $\gcd(N,\, \ell a) = 1$. Any common factor of N and $\ell a \cdot 2^m$ is therefore a power of 2. Let $v = v_2(N)$ and $e = \max(m - v,\, 0)$. In lowest terms, $W = P/d$ with $d = \ell a \cdot 2^e$ and $P \le N$. The product formula gives $\prod_p |\ell|_p = 1/\ell$, and every prime factor of ℓ lies in S, so $\prod_{p \in S} |\ell|_p = 1/\ell$. Since $|a|_p \le 1$,

$$\prod_{p \in S} |d|_p \le \frac{2^{-e}}{\ell}.$$

The left side of the general form is therefore at most

$$d^2 \cdot |\theta - W| \cdot \frac{2^{-e}}{\ell} = \ell a^2\, 2^e \cdot |\theta - W| \le \ell a^2\, 2^m \cdot |\theta - W|.$$

Lemma 2 and $2^m < q + 1$ bound this by $2\ell a^2 (1 + 1/q)(1/y + 2\ell/q)$. Insert $1/y < q^{-1+2\varepsilon}$ and $a < 2(q+1)^{\varepsilon}$. For $q \ge 4\ell$ one has $1 + 1/q < 2$, $1 + 2\ell\, q^{-2\varepsilon} \le 1 + 2\ell$, and $(q+1)^{2\varepsilon} < (2q)^{2\varepsilon}$, so the left side is less than

$$16\ell(1 + 2\ell)\, 2^{2\varepsilon}\, q^{-1+4\varepsilon}.$$

Since $P \le \ell aq + 1$ and $d \le \ell a \cdot 2^m < \ell a(q+1)$, the height satisfies $\max(|P|, d) < \ell a(q+1) + 1 < 4\ell(q+1)^{1+\varepsilon}$, so the right side is greater than $(4\ell)^{-\varepsilon}(q+1)^{-\varepsilon(1+\varepsilon)} > (4\ell)^{-\varepsilon}(2q)^{-\varepsilon(1+\varepsilon)}$. The quotient of these two bounds is

$$16\ell(1+2\ell)\,2^{2\varepsilon}(4\ell)^{\varepsilon}\,2^{\varepsilon(1+\varepsilon)}\,q^{-1+5\varepsilon+\varepsilon^2}.$$

The exponent equals $-1 + 5/(10\ell) + 1/(100\ell^2) \le -1 + 1/4 + 1/400 < -1/2$. The quotient tends to 0 as $q \to \infty$, so Ridout's inequality holds for all q above a bound depending only on ℓ. Then $|\theta - W| \ge \eta$. Lemma 2 and $1/y \le 1$ give $|\theta - W| \le 2(1+2\ell)/q$, hence $q \le (2+4\ell)/\eta$.

In every case q is at most a number depending only on ℓ and η. □

Theorem 2 fails for $\ell = 1$. The Mersenne family $q = 2^e - 1$ gives the cycle $1 \to 1$ for every $e \ge 2$, and the same identity becomes $2^h = q + 1$. The constant η is not explicit, so neither is the bound on q. An effective bound would require an effective inequality of the same shape for every denominator whose odd part is at most a constant times q^{ε}.

4 Cube roots with a fixed odd part

Bauer and Bennett [2] prove an irrationality measure for $2^{1/3}$ and $2^{2/3}$ when the denominator has odd part 1 or 3. The inequality used here is the one obtained in the proof of their Theorem 2.1 from their inequality (5.3), Lemma 4.1 and Proposition 5.1, before they round the exponent to 1.62. The companion paper records it for odd part 1 or 3. The odd part below is an arbitrary fixed positive integer.

Theorem (fixed odd part). Let s be a positive integer and set $s' = s/\gcd(s,3)$. There is an integer $j_0(s') \ge 11000$ such that for $r \in \{1,2\}$, every integer p, and every integer $j \ge j_0(s')$,

$$\left|2^{r/3} - \frac{p}{s\,2^j}\right| > c_3(s')\,2^{-\mu_3 j}.$$

For every $s' \le 16$ one has $j_0(s') = 11000$. For a larger s' the same integer is $11000 + O(\log s')$.

Proof. In the argument of Bauer and Bennett, $s \mid 3$ is used once. It absorbs s into the power of 3 in the denominator of their linear form Λ, and gives $|\Lambda| \ge 1/D$. For a general positive integer s, that denominator gains the factor $s' = s/\gcd(s,3)$, so $|\Lambda| \ge 1/(s'D)$. Their upper bound and their Proposition 5.1 are unchanged, and so is μ_3. The factor $1/s'$ divides the constant, which becomes $c_3(s')$. The choice $r = 2$ in the exponent $N^{-(n+77)/63}$, with $n = 3j + r$, contributes $N^{-79/63}$ after the power of 2^j is removed, and it is the smaller of the two choices $r \in \{1,2\}$.

The term they subtract from the main term is exponentially smaller than $1/D$. It stays smaller than $1/(2s'D)$ for every $j \ge 11000$ when $s' \le 16$. For a larger s' the same comparison holds once $j \ge j_0(s')$, with $j_0(s') = 11000 + O(\log s')$. Both the constant and this integer are read off from their estimates.

Their appeal to the nonvanishing lemma of Beukers (1980/81) uses $s \mid 3$. For a general s, one of the two Padé choices $\delta \in \{0,1\}$ still gives $\Lambda \ne 0$. If both choices vanished, $2^{1/3}$ would be rational. □

Per unit of j the exponent is $\mu_3 \approx 1.61487$, which lies below φ.

5 The bound

The integer $Q(x)$ is defined by the formulas in this section. The comparison that produces them is the proof of Theorem 3.

Length 3. Let x be an odd positive integer. Set

$$A_3 = \frac{x}{2}\, 2^{1/3} \left(\frac{c_3(s'_{\text{near}})}{2^{2/3}\,(1 + 1/x)} \right)^{1/\mu_3}$$

and

$$Q_{3,*}(x) = \left(\frac{2/(3A_3) + 3}{c_3(s'_{\text{win}})\, 2^{1-\mu_3}} \right)^{1/\gamma_3}.$$

Let $J_3(x) = j_0(s'_{\text{near}})$. For each $r \in \{1, 2\}$ and each integer j with $0 \le j < J_3(x)$, and for each odd positive integer y with $y < 3x \cdot 2^{J_3(x)}$, set

$$\rho = \left| \frac{y}{x\, 2^j} \cdot 2^{-r/3} - 1 \right|.$$

Whenever $\rho > 0$, set the corresponding value equal to $(1 + 1/x)/\rho$. The integer $Q_{3,0}(x)$ is the maximum of those finitely many values. Then

$$Q_3(x) = \max(Q_{3,0}(x),\, Q_{3,*}(x)).$$

The factor $x/2$ in A_3 is the condition $|\rho| < 1/2$. The power $2^{1/3}$ is the smaller cube root in the lower bound for another element of the cycle, and $2^{2/3}$ is the larger cube root in the error of that approximation. The numerator $2/(3A_3) + 3$ is two other elements, divided by the length, plus the length.

For every odd $x \le 35$ one has $s'_{\text{near}} \le 35$, and the values of $Q_{3,0}(x)$ below were computed with upper limit 11000 on j. The gate 2^{11001} does not appear in $Q_3(x)$. The quantity $Q_{3,*}(x)$ is smallest at $x = 3$, where $\log_2 Q_{3,*}(3) = 17906.510814$, so $Q_{3,*}(x) > 2^{17906} > 2^{11001}$ for every odd positive integer x.

Length 2. Let $\mu_2 = 1.465$ and $\gamma_2 = 1 - \mu_2 + 1/\mu_2$. The value $\mu_2 = 1.465$ is the exponent printed to three decimals in Corollary 1.4 of Bauer and Bennett [1], for the square root of 2. For each fixed positive integer s that corollary gives an integer $j_0(s)$ such that for every integer p and every integer $j \ge j_0(s)$,

$$\left| \sqrt{2} - \frac{p}{s\, 2^j} \right| > 2^{-\mu_2 j}.$$

The constant in front is 1. The integer $j_0(x)$ is that exponent for $s = x$. Set

$$A_2 = \frac{x}{2}\, 2^{1/2} \left(\frac{1}{2^{1/2}\,(1 + 1/x)} \right)^{1/\mu_2}$$

and

$$Q_{2,*}(x) = \left(\frac{1/(2A_2) + 2}{2^{1-\mu_2}} \right)^{1/\gamma_2}.$$

For each integer j with $0 \le j < j_0(x)$, and for each odd positive integer y with $y < 3x \cdot 2^{j_0(x)}$, set

$$\rho = \left| \frac{y}{x\,2^j} \cdot 2^{-1/2} - 1 \right|.$$

Whenever $\rho > 0$, set the corresponding value equal to $(1 + 1/x)/\rho$. The integer $Q_{2,0}(x)$ is the maximum of those finitely many values. The length-2 bound is this integer:

$$Q_2(x) = Q_{2,0}(x).$$

The quantities $2^{j_0(x)+1}$ and $Q_{2,*}(x)$ are at most $Q_{2,0}(x)$ for every odd positive integer x. The proof of Theorem 3 records that comparison.

Proof of Theorem 3. The statement of Theorem 2 of [5] does not apply: the cycle need not visit 1, and the odd part is x or $3x$ rather than 1 or ℓ. The comparison in its proof does apply, with that odd part.

Consider first a cycle of length 3 through x, written $x \to y \to z \to x$. Lemma 1 gives $2^h = (q + 1/x)(q + 1/y)(q + 1/z)$, so $q = \lfloor 2^{h/3} \rfloor$ and $3 \nmid h$. Write $2^{h/3} = q + \delta$ with $0 < \delta < 1$, and write $h = 3m + r$ with $r \in \{1, 2\}$. The fraction $(3xq + 1)/(3x \cdot 2^m)$ then satisfies

$$\left| 2^{r/3} - \frac{3xq + 1}{3x \cdot 2^m} \right| = \frac{|\delta - 1/(3x)|}{2^m}.$$

Its odd part is $3x$, so Theorem (fixed odd part) applies with $s' = s'_{\text{win}} = x$.

Lemma 1 gives $e^{-3/q} \le 3\delta/\sigma \le e^{3/q}$ with $\sigma = 1/x + 1/y + 1/z$. Therefore $|\delta - \sigma/3| \le (\sigma/q)\, e^{3/q}$, and

$$\left|\delta - 1/(3x)\right| \le (1/y + 1/z)/3 + (\sigma/q)\, e^{3/q}.$$

From $qx + 1 = 2^{k_1} y$ and $2^{h/3} = q + \delta$,

$$\frac{y}{x} = 2^{h/3 - k_1} \cdot \frac{q + 1/x}{q + \delta}.$$

Since $3 \nmid h$, the exponent $h - 3k_1$ is congruent to 1 or 2 modulo 3. Write it as $3j + r'$ with $r' \in \{1, 2\}$ and

$$\frac{y}{x\,2^j} = 2^{r'/3}(1 + \rho), \qquad \rho = \frac{1/x - \delta}{q + \delta}.$$

Then $|\rho| < (1 + 1/x)/q$. The same bound holds for z in place of y. If $|\rho| \ge 1/2$, then $q < 2(1 + 1/x)$. If $|\rho| < 1/2$ and $j \ge J_3(x)$, Theorem (fixed odd part) and $|2^{r'/3} - y/(x\,2^j)| = 2^{r'/3}|\rho| \le 2^{2/3}|\rho|$ give

$$2^j > \left(\frac{c_3(s'_{\text{near}})\, q}{2^{2/3}\,(1 + 1/x)} \right)^{1/\mu_3},$$

and therefore $y > A_3\, q^{1/\mu_3}$. The same lower bound holds for z. Then $\sigma \le 1/x + 2A_3^{-1} q^{-1/\mu_3}$, so

$$(\sigma/q)\, e^{3/q} \le \frac{e^{3/q}}{xq} + 2A_3^{-1} e^{3/q}\, q^{-1-1/\mu_3}.$$

Here $N < 2^{34}$ and $79/63 < 4/3$, so $N^{79/63} < 2^{46}$ and $c_3(s'_{\text{near}}) > 2^{-51}/x$. Also $N > 2^{33}$, so $\mu_3 > 3/2$, hence $1/\mu_3 < 2/3$, and $2^{2/3}(1 + 1/x) \le 4$. Hence $A_3 > 2^{-37}\, x^{1/3}$ and $4A_3^{-1} < 2^{39}$. Combined with $Q_{3,*}(x) > 2^{17906}$, one has $q > 4A_3^{-1}$ whenever $q > Q_{3,*}(x)$. For such q one has $e^{3/q} < 2$ and $q^{-1} \le q^{-1/\mu_3}$, because $\mu_3 > 1$. The first summand is then at most $2\, q^{-1/\mu_3}$, and the second is at most q^{-1/μ_3}. Therefore

$$\bigl|\delta - 1/(3x)\bigr| \le (2/(3A_3) + 3)\, q^{-1/\mu_3}.$$

If $m \ge J_3(x)$ and $\mu_3 > 1$, the lower bound from Theorem (fixed odd part), together with $2^m < q+1$, gives

$$\bigl|\delta - 1/(3x)\bigr| > c_3(s'_{\text{win}})\, 2^{1-\mu_3}\, q^{1-\mu_3}.$$

These two bounds on $|\delta - 1/(3x)|$ conflict once $q > Q_{3,*}(x)$. If $s'_{\text{near}} \le 16$, then $J_3(x) = 11000$, so $q > Q_{3,*}(x)$ forces $q > 2^{11001}$ and $m \ge 11000$. If $s'_{\text{near}} > 16$ and $q \le 2^{J_3(x)+1}$, that power of 2 is an upper bound. It does not change $\max(Q_{3,0}(x), Q_{3,*}(x))$ whenever $J_3(x) \le 17905$, and $J_3(x) = 11000$ for every $s'_{\text{near}} \le 16$.

If instead $0 \le j < J_3(x)$, the value $(1 + 1/x)/\rho$ is one of the numbers whose maximum is $Q_{3,0}(x)$, so $q \le Q_{3,0}(x)$. Thus $q < Q_3(x)$.

Now consider a cycle of length 2 through x, written $x \to y \to x$. Here h is odd. Write $2^{h/2} = q + \delta$ and $h = 2m + 1$. The same comparison, with one other element and with Corollary 1.4 of [1] in place of Theorem (fixed odd part), produces A_2 and $Q_{2,*}(x)$. Both the product around the cycle and the approximation for y have odd part x, and the constant in front of $2^{-\mu_2 j}$ is 1. The numerator $1/(2A_2) + 2$ is that one other element, divided by the length, plus the length.

The exponent $j = 0$ fails the inequality of Corollary 1.4 for every $x \ge 1$, so $j_0(x) \ge 1$. The distance from $x\, 2^{j_0(x)-1}\sqrt{2}$ to an odd integer is at most 1, and the resulting value of $(1 + 1/x)/\rho$ is at least $(x + 1)\sqrt{2}\, 2^{j_0(x)-1}$. For $x \ge 3$ this is at least $2^{j_0(x)+1}$, because $(x + 1)\sqrt{2} \ge 4\sqrt{2} > 4$. For $x = 1$ one has $j_0(1) = 9$, and the value at $j = 7$ has logarithm $14.192568 > 10$. Thus $2^{j_0(x)+1} \le Q_{2,0}(x)$.

The quantity A_2 increases with x, so $Q_{2,*}(x)$ decreases, and $Q_{2,*}(1) < 2^{11}$. Every exponent j with $x \cdot 2^{-0.465j} \ge 1/2$ fails the inequality, and the largest such j contributes at least $(1 + 1/x)\, x\sqrt{2}\, 2^j$ even when the distance to an odd integer is 1. For every odd $x \ge 5$ that lower bound already exceeds $Q_{2,*}(x)$. For $x = 1$ and $x = 3$ the values $Q_{2,0}(1) > 2^{14}$ and $Q_{2,0}(3) > 2^{17}$ do the same. Thus $Q_{2,*}(x) \le Q_{2,0}(x)$, and $Q_2(x) = Q_{2,0}(x)$, and a length-2 cycle satisfies $q < Q_2(x)$.

For $x = 1$ and length 2, Proposition A.1 of [5] gives the only such q, namely $q = 5$. That conclusion does not change the formula for $Q_2(1)$. □

The value at $x = 1$, length 3. Here $s'_{\text{near}} = 1$ and $s'_{\text{win}} = 1$, so $J_3(1) = 11000$. The maximum $Q_{3,0}(1)$ is less than 2^{11003}, and $Q_{3,*}(1)$ is less than 2^{17990}, so

$$Q_3(1) = Q_{3,*}(1) < 2^{17990}.$$

Each h gives one candidate $q = \lfloor 2^{h/3} \rfloor$. The bound $q < 2^{17990}$ means it is enough to check $h < 53970$. The companion paper checks every $h \le 200{,}000$ and finds no cycle, so that search includes the range. Theorem 1 of [5] is this case.

The proof of that theorem in the companion paper reaches the larger cutoff $q < 2^{22003}$. There the symbol x is the element after 1. The argument asks for that element to clear 2^{11001}, using a lower bound of the shape $\sqrt{q/2}$. The integer $Q_3(1)$ does not use that lower bound. Once $q > Q_{3,0}(1)$, an exponent below 11000 is already impossible, and the conflict is $Q_{3,*}(1)$.

Powers of two above $Q_{3,*}(x)$. The integer $\lceil Q_{3,*}(x) \rceil$ is the sharper bound, and it has thousands of digits. The table uses the next power of 2 above $Q_{3,*}(x)$, read off from the logarithm. For $x = 5$ one has $\log_2 Q_{3,*}(5) = 18213.724287504393$, so $\lceil Q_{3,*}(5) \rceil$ has 5483 digits and begins 753951296..., while the table entry is 2^{18214}. The same rounding, $2^{\lceil \log_2 \cdot \rceil}$, is used for $Q_{3,0}(x)$, $Q_{2,*}(x)$ and $Q_2(x)$.

For every odd x from 1 to 35, the entry $2^{\lceil \log_2 Q_{3,*}(x) \rceil}$ is:

x		x		x	
1	2^{17990}	13	2^{18387}	25	2^{18514}
3	2^{17907}	15	2^{18190}	27	2^{18304}
5	2^{18214}	17	2^{18439}	29	2^{18543}
7	2^{18273}	19	2^{18460}	31	2^{18556}
9	2^{18094}	21	2^{18255}	33	2^{18344}
11	2^{18356}	23	2^{18497}	35	2^{18580}

Rows in which 3 divides x sit below their neighbors because $s'_{\text{near}} = x/\gcd(x, 3)$. The row $x = 3$ has $s'_{\text{near}} = 1$ and $s'_{\text{win}} = 3$, so the factor 3 in $c_3(s'_{\text{win}})$ is the place this table uses Theorem (fixed odd part) past odd part 1 or 3.

Powers of two above $Q_{3,0}(x)$. For the same odd x, with the maximum taken over $0 \le j < 11000$, the entry $2^{\lceil \log_2 Q_{3,0}(x) \rceil}$ is:

x		x		x	
1	2^{11003}	13	2^{11007}	25	2^{11009}
3	2^{11002}	15	2^{11006}	27	2^{11006}
5	2^{11006}	17	2^{11009}	29	2^{11012}
7	2^{11006}	19	2^{11009}	31	2^{11008}
9	2^{11006}	21	2^{11006}	33	2^{11005}
11	2^{11005}	23	2^{11006}	35	2^{11008}

In every row the exponent j that attains the maximum lies in $\{10997, 10998, 10999\}$. For these eighteen values, $Q_{3,0}(x) < Q_{3,*}(x)$. That comparison is checked for odd $x \le 35$. It is not proved for every odd x. From $t^3 - 2^r$, if the distance from $s'_{\text{near}}\, 2^{j+r/3}$ to an integer is positive, that distance is greater than $1/(Q^2 \cdot 3(2^{2/3} + 1)^2)$ once the integer is within 1 of $s'_{\text{near}}\, 2^{j+r/3}$, with $Q = s'_{\text{near}} 2^j$. With $j \le 10999$ and $s'_{\text{near}} \le x$ this only yields $Q_{3,0}(x) < x^3\, 2^{33003}$. At $x = 1$ the two logarithms, of this upper estimate and of $Q_{3,*}(1)$, are 33003 and 17989. The estimate $x^3\, 2^{33003}$ stays above $Q_{3,*}(x)$ until x is about 2^{67}. Past that size, $Q_{3,*}(x)$ grows like x^{1/γ_3} with $1/\gamma_3 \approx 229$, while $x^3\, 2^{33003}$ grows like x^3.

Powers of two above $Q_{2,*}(x)$. This integer is used in the proof and is at most $Q_2(x)$. As $x \to \infty$, $\log_2 Q_{2,*}(x)$ decreases to $\mu_2/\gamma_2 \approx 6.733$. The row $x = 11$ has logarithm 7.013, so that one ceiling moves if the fourth digit of the exponent 1.465 is lower. For odd x from 1 to 35, the entry

$2^{\lceil \log_2 Q_{2,*}(x) \rceil}$ is 2^{11} at $x = 1$, then 2^8 for $x = 3, 5, 7, 9, 11$, and 2^7 for every odd x from 13 to 35.

The integer $Q_2(x)$. For these eighteen values the last failure of $|\sqrt{2} - p/(x\,2^j)| > 2^{-1.465j}$ lies at $j \le 21$, and a check through $j < 53620$ finds no later failure. The entry is $2^{\lceil \log_2 Q_2(x) \rceil}$.

x	$j_0(x)$	winning j	$Q_2(x)$ below
1	9	7	2^{15}
3	12	11	2^{18}
5	22	17	2^{33}
7	12	10	2^{18}
9	12	11	2^{18}
11	13	12	2^{19}
13	14	9	2^{21}
15	22	17	2^{32}
17	15	13	2^{23}
19	17	14	2^{25}
21	17	12	2^{25}
23	16	14	2^{23}
25	22	17	2^{32}
27	15	14	2^{22}
29	17	16	2^{25}
31	17	16	2^{25}
33	15	13	2^{22}
35	22	17	2^{32}

The cycle $27 \to 611 \to 27$ has $q = 181 < 2^{22}$. The cycle $35 \to 99 \to 35$ has $q = 181 < 2^{32}$.

6 Cycles of length 2

Throughout this section, $x \to y \to x$ is a cycle of length 2, and x is its smallest element. The steps are $qx + 1 = 2^{k_1} y$ and $qy + 1 = 2^{k_2} x$, and $h = k_1 + k_2$. Lemma 1 gives $q = \lfloor 2^{h/2} \rfloor$ and h odd. Let $D = 2^h - q^2$.

Lemma 3 (the length-2 equation). Then $k_1 < k_2$. Let $t = k_2 - k_1$. Then t is positive and odd, $h = 2k_1 + t$, and

$$x = \frac{q + 2^{k_1}}{D}, \qquad y = \frac{q + 2^{k_2}}{D}.$$

The integer D divides $2^t - 1$, and $D \equiv 7 \pmod 8$. Every prime divisor of D is congruent to 1 or 7 modulo 8. Let $u = (2^t - 1)/D$. Then u is a positive odd integer,

$$y = x + u\,2^{k_1}, \qquad y^2 - 2^t x^2 = u,$$

$2^{k_1} \le q - 1$, $t > \log_2(q/x)$, and $y^2 > qx + 1$.

Proof. Since h is odd, $k_1 \ne k_2$. From $qy + 1 = 2^{k_2} x$ and $y = (qx + 1)/2^{k_1}$,

$$2^h x = q(qx + 1) + 2^{k_1},$$

so $x(2^h - q^2) = q + 2^{k_1}$. Lemma 1 gives $q^2 < 2^h$, so $D > 0$ and $x = (q + 2^{k_1})/D$. The same elimination with the roles of x and y exchanged gives $y = (q + 2^{k_2})/D$. Since $x < y$ and $D > 0$, one has $k_1 < k_2$. Thus $t > 0$. From $h = 2k_1 + t$ and h odd, t is odd.

Now D divides $q + 2^{k_1}$ and $q + 2^{k_2}$, so D divides

$$(q + 2^{k_2}) - (q + 2^{k_1}) = 2^{k_1}(2^t - 1).$$

Here q is odd and $h \geq 3$, so 2^h is divisible by 8 and $q^2 \equiv 1 \pmod 8$. Thus $D \equiv 7 \pmod 8$, and in particular D is odd. Hence D divides $2^t - 1$. Let p be a prime divisor of D. Then $q^2 = 2^h - D \equiv 2^h \pmod p$. If p divides q, then p divides 2^h. But D is odd, so $p \neq 2$, and p does not divide 2^h. Thus 2^h is a nonzero square modulo p. Since h is odd, 2 is a quadratic residue modulo p, and therefore $p \equiv 1 \pmod 8$ or $p \equiv 7 \pmod 8$.

The quotient $u = (2^t - 1)/D$ is a positive integer. Both $2^t - 1$ and D are odd, so u is odd. Then

$$y - x = \frac{2^{k_2} - 2^{k_1}}{D} = u\, 2^{k_1},$$

so $y = x + u\, 2^{k_1}$. Substitute $q = Dx - 2^{k_1}$ into $qx + 1 = 2^{k_1} y$:

$$Dx^2 - 2^{k_1+1}x + 1 = u\, 4^{k_1}.$$

Therefore

$$\begin{aligned} y^2 - 2^t x^2 &= \left(x + u\, 2^{k_1}\right)^2 - 2^t x^2 \\ &= u\, 2^{k_1+1}x + u^2 4^{k_1} - x^2(2^t - 1) \\ &= u\big(2^{k_1+1}x + u\, 4^{k_1} - Dx^2\big) \\ &= u. \end{aligned}$$

Since x and y are distinct and odd, $y \geq x + 2$. Then

$$2^{k_1} = \frac{qx + 1}{y} \leq \frac{qx + 1}{x + 2}.$$

The inequality $(qx + 1)/(x + 2) < q$ is $1 < 2q$. Thus $2^{k_1} < q$. Both are integers, so $2^{k_1} \leq q - 1$.

From $y - x \geq 2^{k_1}$ and $2^{k_1} = (qx + 1)/y$,

$$2^{k_1}\big(x + 2^{k_1}\big) \leq qx + 1,$$

so $2^{2k_1} \leq x(q - 2^{k_1}) + 1$. For $k_1 \geq 1$ the right side is at most $x(q - 2) + 1$, and $x(q - 2) + 1 < xq$ because $1 < 2x$. Thus $2^{2k_1} < xq$. Lemma 1 gives $2^{h/2} = q + \delta$ with $\delta > 0$, so $h > 2\log_2 q$. Therefore

$$t = h - 2k_1 > 2\log_2 q - \log_2(xq) = \log_2(q/x).$$

Finally $y(y - x) \geq y \cdot 2^{k_1} = qx + 1$. Since $x \geq 1$, one has $y > y - x$, and therefore $y^2 > qx + 1$. $\square$

Lemma 4 (the case $D = 7$). If $D = 7$, the cycle is $1 \to 3$ with $q = 5$, or $35 \to 99$ with $q = 181$, or $27 \to 611$ with $q = 181$.

Proof. Here $q^2 + 7 = 2^h$. By Theorem 1.8 of Bauer and Bennett (2002), the positive solutions (q, h) are $(1, 3)$, $(3, 4)$, $(5, 5)$, $(11, 7)$ and $(181, 15)$. Lemma 3 requires $q \geq 3$ and h odd, which leaves $(5, 5)$, $(11, 7)$ and $(181, 15)$. The order of 2 modulo 7 is 3, so 3 divides t, and Lemma 3 requires t odd and $1 \leq t \leq h - 2$.

For $(q, h) = (5, 5)$ the only such t is 3, so $k_1 = 1$ and $k_2 = 4$. Then $(5+2^1)/7 = 1$ and $(5+2^4)/7 = 3$, which is $1 \to 3$.

For $(q, h) = (11, 7)$ one has $-11 \equiv 3 \pmod 7$. The powers of 2 modulo 7 are 2, 4 and 1, so $q + 2^{k_1}$ is never divisible by 7.

For $(q, h) = (181, 15)$ the admissible values are $t = 3$ and $t = 9$. If $t = 3$, then $k_1 = 6$ and $k_2 = 9$, and

$$\frac{181 + 2^6}{7} = 35, \qquad \frac{181 + 2^9}{7} = 99,$$

which is $35 \to 99$. If $t = 9$, then $k_1 = 3$ and $k_2 = 12$, and

$$\frac{181 + 2^3}{7} = 27, \qquad \frac{181 + 2^{12}}{7} = 611,$$

which is $27 \to 611$. □

Lemma 5 (the case $u = 1$). If $u = 1$, the cycle is $1 \to 3$ with $q = 5$, or $35 \to 99$ with $q = 181$.

Proof. Here $D = 2^t - 1$, so $q^2 = 2^{2k_1+t} - 2^t + 1$ and

$$(q - 1)(q + 1) = 2^t(2^{2k_1} - 1).$$

The factor $2^{2k_1} - 1$ is odd, and $\gcd(q - 1, q + 1) = 2$, so

$$v_2(q - 1) + v_2(q + 1) = t$$

with the smaller valuation equal to 1. Lemma 3 gives t odd and $D \equiv 7 \pmod 8$, so $t \geq 3$. Let $c = t - 2$, which is odd and at least 1.

If $v_2(q + 1) = 1$ and $v_2(q - 1) = t - 1$, write $q + 1 = 2r$ and $q - 1 = 2^{t-1}s$ with r and s odd. Then $r = 2^{t-2}s + 1$ and $s(2^{t-2}s + 1) = 2^{2k_1} - 1$, so

$$2^c s^2 + s + 1 = 2^{2k_1}.$$

If $v_2(q - 1) = 1$ and $v_2(q + 1) = t - 1$, write $q - 1 = 2s$ and $q + 1 = 2^{t-1}r$ with r and s odd. Then $s = 2^{t-2}r - 1$ and $r(2^{t-2}r - 1) = 2^{2k_1} - 1$, so

$$2^c r^2 - r + 1 = 2^{2k_1}.$$

In either case, for a positive odd integer X,

$$\left(2^{c+1}X \pm 1\right)^2 + (2^{c+2} - 1) = 2^{2k_1+c+2},$$

the plus sign for $X = s$ and the minus sign for $X = r$.

Suppose $c \geq 3$. Then $2^{c+2} - 1 = 2^k - 1$ with $k = c + 2 \geq 5$. Theorem 1.8 of Bauer and Bennett (2002) gives exactly two positive solutions of $Y^2 + (2^k - 1) = 2^n$, namely $Y = 1$ with $n = k$, and $Y = 2^{k-1} - 1$ with $n = 2k - 2$. The first gives $2k_1 + c + 2 = c + 2$, so $k_1 = 0$. The second gives $2k_1 + c + 2 = 2c + 2$, so $2k_1 = c$. But c is odd, so neither solution has an integer $k_1 \geq 1$.

Now $c = 1$, so $t = 3$ and $D = 7$, and the equation is $Y^2 + 7 = 2^{2k_1+3}$ with $Y = 4X \pm 1$. The same theorem gives $Y \in \{1, 3, 5, 11, 181\}$ and $2k_1 + 3 \in \{3, 4, 5, 7, 15\}$. The exponent $2k_1 + 3$ is odd and at least 5, which leaves 5, 7 and 15. If the exponent is 5, then $Y = 5$ and $k_1 = 1$. The equation $4X - 1 = 5$ has no integer solution, while $4X + 1 = 5$ gives $X = 1$. This is the plus sign, $q - 1 = 2^2 \cdot 1$, and $q = 5$, the cycle $1 \to 3$. If the exponent is 7, then $Y = 11$ and $k_1 = 2$. The equation $4X + 1 = 11$ has no integer solution, while $4X - 1 = 11$ gives $X = 3$. This is the minus sign and $q = 11$. Then $(11 + 2^2)/7 = 15/7$, which is not an integer. If the exponent is 15, then $Y = 181$ and $k_1 = 6$. The equation $4X - 1 = 181$ has no integer solution, while $4X + 1 = 181$ gives $X = 45$. This is the plus sign, $q - 1 = 2^2 \cdot 45$, and $q = 181$, the cycle $35 \to 99$. □

Proposition 5. Any cycle of length 2 other than $1 \to 3$, $35 \to 99$ and $27 \to 611$ has

$$q \geq 2^{4500}, \qquad D \geq 15, \qquad u \geq 3,$$

every odd element at least 37, and, writing $h = 2m + 1$,

$$m \geq 4500, \qquad D > 2^{0.535m+1}, \qquad t \geq 2409.$$

In particular $k_1 \leq \lfloor \log_2 q \rfloor - 1204$ and $k_2 \geq \lfloor \log_2 q \rfloor + 1205$. Also $t > \log_2(3q/x)$.

Proof. The companion paper searches every cycle of length 2 with $h \leq 9000$ and finds only the three cycles named above. That search excludes every cycle of length 2 with $q < 2^{4500}$. Thus any other cycle has $q \geq 2^{4500}$. Since $2^{h/2} > q$, one has $h > 2 \cdot 4500$, so $h \geq 9001$. With $h = 2m + 1$ this is $m \geq 4500$.

Lemma 4 gives $D \neq 7$. Lemma 3 gives $D \equiv 7 \pmod 8$ and $D > 0$, so $D \geq 15$. Lemma 5 gives $u \neq 1$. Lemma 3 gives that u is a positive odd integer, so $u \geq 3$.

Let z be an odd element of the cycle with $z \leq 35$. Theorem 3 gives $q < Q_2(z)$. Section 5 records $Q_2(z) \leq 2^{33}$ for every odd $z \leq 35$. Then $q < 2^{33} < 2^{4500}$, so the cycle is one of the three found by the search. Each of those has its odd elements in $\{1, 3, 27, 35, 99, 611\}$. An element $z \leq 35$ on one of them is 1, 3, 27 or 35. Therefore every odd element of any other cycle is at least 37.

Corollary 1.4 of Bauer and Bennett (2002), applied with odd part 1, gives

$$\left| \sqrt{2} - \frac{q}{2^m} \right| > 2^{-1.465m}$$

for every $m \geq j_0(1)$. Section 5 records $j_0(1) = 9$, and here $m \geq 4500$. Writing $2^{h/2} = q + \delta$ as in Lemma 1,

$$\delta > 2^{-0.465m}.$$

Then $D = \delta(2q + \delta) > 2q\delta$, so

$$\log_2 D > 1 + \log_2 q - 0.465m.$$

Since $q > 2^m\sqrt{2} - 1$ and $m \geq 4500$,

$$\log_2 q > m + \tfrac{1}{2} - 2^{-4499}.$$

Hence $\log_2 D > 0.535m + 1.5 - 2^{-4499} > 0.535m + 1$, and $D > 2^{0.535m+1}$. At $m = 4500$ this is $D > 2^{2408.5}$. For $m > 4500$ the same lower bound is larger. Lemma 3 gives $D \leq 2^t - 1$, so $t \geq 2409$.

Since t is odd, $(t-1)/2$ is an integer, and $k_1 = m-(t-1)/2$. Thus $k_1 \leq m-1204$. Since $m \geq 4500$ and $\sqrt{2} - 1 > 2^{-m}$, one has $2^m\sqrt{2} > 2^m + 1$, so $q \geq 2^m + 1$. Also $\sqrt{2} < 2$, so $q < 2^m\sqrt{2} < 2^{m+1}$. Thus $\lfloor \log_2 q \rfloor = m$ and $k_1 \leq \lfloor \log_2 q \rfloor - 1204$. Also $k_2 = m + (t+1)/2 \geq m + 1205 = \lfloor \log_2 q \rfloor + 1205$.

For the last claim, $u \geq 3$ gives $y - x \geq 3 \cdot 2^{k_1}$. Combined with $2^{k_1} = (qx + 1)/y$,

$$3 \cdot 4^{k_1} \leq x(q - 2^{k_1}) + 1 < xq,$$

so $2^{2k_1} < xq/3$. With $h > 2\log_2 q$,

$$t = h - 2k_1 > 2\log_2 q - \log_2(xq/3) = \log_2(3q/x).$$

□

The case that remains is $u \geq 3$. Proposition 5 puts $q \geq 2^{4500}$ and $t \geq 2409$, with D dividing $2^t - 1$ and $(q + 2^{(h-t)/2})/D$ an odd integer at least 37.

7 Longer cycles

Length 3 is the longest length for which every cycle through a fixed odd x falls under a root controlled by Theorem (fixed odd part) or by Corollary 1.4 of [1]. Two thinner classes go further. They depend on ℓ and on h modulo ℓ, so they are a corollary rather than part of Theorem 3.

Corollary 4. Let x_0 be an odd positive integer. There is an integer $Q(x_0, \ell)$, read off from x_0 and ℓ by the comparison in the proof of Theorem 3, with the following property. If x_0 lies on a cycle of length ℓ with halving total h, and either

1. $\ell = 2^a$ with $a \geq 2$ and $h \equiv \ell/2 \pmod{\ell}$, or

2. $\ell = 3 \cdot 2^a$ with $a \geq 1$ and $h \equiv \ell/3$ or $2\ell/3 \pmod{\ell}$,

then $q < Q(x_0, \ell)$.

Length 4 with h odd is not included. The first family uses Corollary 1.4 of [1]. The second uses Theorem (fixed odd part).

Proof. In the first family every irrational root that occurs is $\sqrt{2}$. An element aimed at an integer power of 2 is large because it is odd. Corollary 1.4 of [1] supplies the exponent $\mu_2 = 1.465 < \varphi$

with odd part x_0. The comparison in the proof of Theorem 3 then produces a finite check of the exponents below $j_0(x_0)$ and an explicit conflict for the larger exponents.

In the second family the only irrational roots that occur are $2^{1/3}$ and $2^{2/3}$. Length 3 is the case $a = 0$, in which those two residues are every nonzero class modulo 3, and the restriction on h disappears. For $a \geq 1$, Theorem (fixed odd part) supplies the same exponent $\mu_3 < \varphi$ with odd part x_0 or $3x_0$. The comparison in the proof of Theorem 3 again produces the integer $Q(x_0, \ell)$. □

Corollaries 12, 13 and 14 of [5] are the case $x_0 = 1$. Corollary 12 of that paper, and Theorem 1 of that paper for length 3, conclude that no such q exists. Corollary 4 gives a bound in terms of x_0. For $x_0 = 1$ the search that turns the length-3 bound into emptiness is already done in the companion paper. For a general x_0 the bound $Q(x_0, \ell)$ does not by itself rule the cycle out. Corollaries 13 and 14 of [5] keep the constants of those papers ($c = 1$ and $j_0 = 9$ for $\sqrt{2}$; $j_0 = 11000$ for the cube roots). The integer $Q(x_0, \ell)$ still depends on x_0, so it gives no uniform bound on every cycle of that length and residue.

8 A heuristic

A cycle of length $\ell \geq 2$ with h halvings has to satisfy several conditions at once. As above, $q^\ell < 2^h < (q+1)^\ell$, so ℓ does not divide h and $q = \lfloor 2^{h/\ell} \rfloor$ is a single odd integer at least 3. Write $D = 2^h - q^\ell$. Starting from any element x of the cycle, the halving counts are a composition of h into ℓ positive parts, and the partial sums S_j begin at x. Unfolding the cycle relations gives

$$x = \frac{N}{D}, \qquad N := q^{\ell-1} + \sum_{j=1}^{\ell-1} q^{\ell-1-j}\, 2^{S_j}.$$

Thus D divides N, and x is a positive odd integer. Tracing from x recovers those counts exactly, and the ℓ elements are odd and distinct.

If, for a random composition, N is uniformly distributed modulo D, then D divides N with probability $1/D$. Counting cycles up to rotation, the expected number at this pair (h, ℓ) is then about

$$\frac{1}{\ell D} \binom{h-1}{\ell-1}.$$

This is only a heuristic: N is a sparse combination of powers of 2 with coefficients powers of q, so the residue N mod D can be constrained, and the remaining conditions only make a cycle rarer.

The estimate is large only when D is small, that is when q^ℓ is extraordinarily close to a power of 2. The known cycles are exactly those coincidences:

$$5^2 = 2^5 - 7, \qquad 5^3 = 2^7 - 3, \qquad 181^2 = 2^{15} - 7.$$

The corresponding expectations are $4/(2 \cdot 7) \approx 0.29$, $15/(3 \cdot 3) = 5/3$, and $14/(2 \cdot 7) = 1$, against one, two, and two cycles found. The only consecutive perfect powers are 8 and 9, of the other sign, so $D \geq 3$ for $\ell \geq 2$, and $D = 3$ and $D = 7$ are as small as occur here.

The next-smallest values of D do not produce cycles. For $q = 11$ and $h = 7$ one has $11^2 = 2^7 - 7$, the same D as for $q = 5$ and $q = 181$, and the estimate is $6/(2 \cdot 7) \approx 0.43$; but $11 + 2^{k_1} \not\equiv 0 \pmod 7$

for every positive $k_1 < 7$, so D never divides N. For $q = 45$ and $h = 11$, likewise $45^2 = 2^{11} - 23$ and $45 + 2^{k_1} \equiv 0 \pmod{23}$ forces $2^{k_1} \equiv 1 \pmod{23}$, hence $11 \mid k_1$, which is impossible for $0 < k_1 < 11$. For $q = 3$ and $h = 5$, $3^3 = 2^5 - 5$ and the estimate is 0.4; the length-3 search finds no cycle. For $q = 3$ and $h = 8$, $3^5 = 2^8 - 13$ and the estimate is 0.54; the length-5 search finds no cycle. For $q = 19$ and $h = 17$, $19^4 = 2^{17} - 751$, the fourth-power near miss recorded in the companion paper [5], and the estimate is 0.19; the length-4 search finds no cycle.

Summing the same estimate over every admissible pair with $2 \le \ell \le 20$ and $h \le 200$ gives a total of about 7 expected cycles, of which about 3 sit in the three known slots and the remaining 4 are spread over pairs that the searches in the companion paper [5] have already emptied. For large h a typical D is of size $\ell\delta\, 2^{h(1-1/\ell)}$, with δ the fractional part of $2^{h/\ell}$, and $\binom{h-1}{\ell-1}/D$ decays exponentially in h. Length 1 is different: $D = 2^h - q = 1$ holds for every Mersenne q, and $N = 1$, so the division succeeds for every e and produces $1 \to 1$.

For a cycle at large q there is a further obstruction. A random odd input has $E[v_2(qn+1)] = 2$, while a cycle needs mean $h/\ell \approx \log_2 q$. For $q = 5$ this is 2.32, a mild excess, and three cycles exist. For $q = 181$ it is 7.5, which occurs only because $D = 7$. For larger q the same mean requires a still smaller relative gap, which is the Diophantine condition already used in Theorem 2 of the companion paper and Theorem 2.

9 Open problems

1. **All cycles.** Conjecture 1 is open. Theorem 2 gives, for each $\ell \ge 2$, only finitely many q with a cycle of length ℓ, with no bound and no classification. A bound in terms of a fixed odd element, for lengths 2 and 3, is Theorem 3, and for the classes in Corollary 4 it is that corollary. A uniform bound on q remains open. For length 2, Section 6 leaves the case $u \ge 3$. The heuristic of Section 8 is not a proof, and the searches leave infinitely many pairs (ℓ, h) unchecked.

2. **Novelty.** Franco's 1990 thesis [4] on Diophantine approximation and the $qx + 1$ problem could not be consulted and should be checked for overlap, including with Theorem 2.

References

[1] M. Bauer and M. A. Bennett, Applications of the hypergeometric method to the generalized Ramanujan–Nagell equation, Ramanujan J. 6 (2002), 209–270, doi:10.1023/A:1015779301077.

[2] M. Bauer and M. A. Bennett, Ramanujan–Nagell cubics, Rocky Mountain J. Math. 48 (2018), 385–412, doi:10.1216/rmj-2018-48-2-385.

[3] F. Beukers, On the generalized Ramanujan–Nagell equation I, Acta Arith. 38 (1980/81), 389–410, doi:10.4064/aa-38-4-389-410.

[4] Z. Franco, Diophantine Approximation and the $qx+1$ Problem, PhD thesis, UC Berkeley, 1990.

[5] T. Hackman, I. Cycles through 1 for the $qn + 1$ map: no cycles with three odd steps, and constraints on longer cycles, companion paper.

[6] D. Ridout, The p-adic generalization of the Thue–Siegel–Roth theorem, Mathematika 5 (1958), 40–48, doi:10.1112/S0025579300001339.

[7] R. Santos, On the Collatz general problem $qn + 1$, preprint (2020), arXiv:2005.00346.